Environ Health Perspect

DOI: 10.1289/EHP9620

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Supplemental Material

Exposure to Outdoor Particulate Matter Air Pollution and Risk of Gastrointestinal Cancers in Adults: A Systematic Review and Meta-Analysis of Epidemiologic Evidence

Natalie Pritchett, Emily C. Spangler, George M. Gray, Alicia A. Livinski, Joshua N. Sampson, Sanford M. Dawsey, and Rena R. Jones

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Figure S6. Funnel plot of studies of the association between $PM_{2.5}$ and gastric cancer.

Additional File- Excel Document

Table S1. PRISMA 2020 item checklist

Section and topic	Item #	Checklist item	Location where item is reported
Title			
Title	1	Identify the report as a systematic review.	Page 1, lines 1-2
Abstract			
Abstract	2	See the PRISMA 2020 for Abstracts checklist (Table 2).	Pages 3-4, lines 27-59
Introduction			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Pages 6-7, lines 95-114
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 7, lines 1116-121
Methods			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 8-9, lines 150-170
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 8, lines 137-141
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 8, lines 141-149 Table 1, Table S2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 9, lines 160-170
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 9, lines 172-175
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 9-10, lines 179-183
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 9, lines 175-178
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Pages 10-11, lines 185-218
Effect measures	12	Specify for each outcome the effect measure(s) (e.g., risk ratio, mean difference) used in the synthesis or presentation of results.	Page 12, lines 229-232
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 12, lines 240-242
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 12, lines 232-235
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 12, lines 236-237
	13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 12, lines 236-248
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g., subgroup analysis, metaregression).	Page 12, lines 246-248
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results.	Page 13, lines 254-260
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Pages 10-11, lines 200-218

(*continued on next page*)

Supplemental Table 1 (*continued*)

Section and topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 11, lines 219-225
Results			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (see Fig. 1).	Page 13, lines 263-267
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 13, lines 267-270; Table S3
Study characteristics	17	Cite each included study and present its characteristics.	Page 14, lines 271-275
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Pages 14-16, lines 282-331
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g., confidence/credible interval), ideally using structured tables or plots.	Table 4
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Figure 2 and Table S4
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g., confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Pages 17-18, lines 352-363; Figure 3
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Pages 17-18, lines 352-363; Figure 3
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results.	Page 18, lines 364-378
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 17, lines 337-341
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Pages 18-19, lines 379-391
Discussion			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 19, lines 392-405
	23b	Discuss any limitations of the evidence included in the review.	Pages 20-23, lines 423-497
	23c	Discuss any limitations of the review processes used.	Page 24, lines 498-508
	23d	Discuss implications of the results for practice, policy, and future research.	Pages 24-25, lines 509-527
Other information			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 7, line 127
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 7, line 126
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or nonfinancial support for the review, and the role of the funders or sponsors in the review.	Page 25, line 537
Competing interests	26	Declare any competing interests of review authors.	Page 25, lines 539-540
Availability of data, code, and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Tables 3, 4, S3, S4

Section and topic	Item #	Checklist item
Title		
Title	1	Identify the report as a systematic review.
Background		
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.
Methods		
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.
Synthesis of results	6	Specify the methods used to present and synthesise results.
Results		
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).
Discussion		
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g., study risk of bias, inconsistency and imprecision).
Interpretation	10	Provide a general interpretation of the results and important implications.
Other		
Funding	11	Specify the primary source of funding for the review.
Registration	12	Provide the register name and registration number.

* This abstract checklist retains the same items as those included in the PRISMA for Abstracts statement published in 2013 [54], but has been revised to make the wording consistent with the PRISMA 2020 statement and includes a new item recommending authors specify the methods used to present and synthesise results (item #6).

Table S2. Final search strategies used for literature searches

Database: PubMed
Vendor: US National Library of Medicine
Date of searches: July 2, 2019, March 15, 2021, September 7, 2021
Limits used: English language & publication date: 01/01/1980 – 12/31/2019; 01/01/2019 – 12/31/2020; 01/01/2020 – 12/31/2021

Concept	Search Terms Used
Gastrointestinal Cancer	("Esophageal cancer"[tiab] OR "Esophageal cancers"[tiab] OR "oesophageal cancer"[tiab] OR "oesophageal cancers"[tiab] OR "Gastric cancer"[tiab] OR "Gastric cancers"[tiab] OR "esophageal adenocarcinoma"[tiab] OR "esophageal adenocarcinomas"[tiab] OR "oesophageal adenocarcinoma"[tiab] OR "oesophageal adenocarcinomas"[tiab] OR "Upper aerodigestive tract cancer"[tiab] OR "Upper aerodigestive tract cancers"[tiab] OR "Stomach cancer"[tiab] OR "Stomach cancers"[tiab] OR "esophageal squamous cell carcinoma"[tiab] OR "esophageal squamous cell carcinomas"[tiab] OR "oesophageal squamous cell carcinoma"[tiab] OR "oesophageal squamous cell carcinomas"[tiab] OR "Upper gastrointestinal cancer"[tiab] OR "Upper gastrointestinal cancers"[tiab] OR "Esophageal neoplasm"[tiab] OR "Esophageal neoplasms"[tiab] OR "oesophageal neoplasm"[tiab] OR "oesophageal neoplasms"[tiab] OR "Gastric neoplasm"[tiab] OR "Gastric neoplasms"[tiab] OR "esophageal adenocarcinoma"[tiab] OR "esophageal adenocarcinomas"[tiab] OR "oesophageal adenocarcinoma"[tiab] OR "oesophageal adenocarcinomas"[tiab] OR "Upper aerodigestive tract neoplasms"[tiab] OR "Stomach neoplasm"[tiab] OR "Stomach neoplasms"[tiab] OR "esophageal squamous cell carcinoma"[tiab] OR "esophageal squamous cell carcinomas"[tiab] OR "oesophageal squamous cell carcinoma"[tiab] OR "oesophageal squamous cell carcinomas"[tiab] OR "Upper gastrointestinal neoplasm"[tiab] OR "Upper gastrointestinal neoplasms"[tiab] OR "alimentary carcinoma"[tiab] OR "gastrointestinal cancer"[tiab] OR "gastrointestinal cancers"[tiab] OR "gastrointestinal neoplasm"[tiab] OR "gastrointestinal neoplasms"[tiab] OR "Gastrointestinal Tract Cancer"[tiab] OR "Gastrointestinal Tract Cancers"[tiab] OR "Gastrointestinal Neoplasms"[tiab] OR "liver cancer"[tiab] OR "liver cancers"[tiab] OR "hepatic neoplasms"[tiab] OR "hepatic neoplasm"[tiab] OR "liver neoplasm"[tiab] OR "liver neoplasms"[tiab] OR "hepatic cancer"[tiab] OR "hepatic cancers"[tiab] OR "hepatic neoplasm"[tiab] OR "hepatic neoplasms"[tiab] OR "hepatocellular cancer"[tiab] OR "hepatocellular cancers"[tiab] OR "hepatocellular neoplasm"[tiab] OR "hepatocellular neoplasms"[tiab] OR cholangiocarcinoma OR cholangiocarcinomas OR "cholangiocellular carcinoma"[tiab] OR "cholangiocellular carcinomas"[tiab] OR "extrahepatic cholangiocarcinoma"[tiab] OR "extrahepatic cholangiocarcinomas"[tiab] OR "intrahepatic cholangiocarcinoma"[tiab] OR "intrahepatic cholangiocarcinomas"[tiab] OR "pancreatic cancer"[tiab] OR "pancreatic cancers"[tiab] OR "pancreatic neoplasm"[tiab] OR "pancreatic neoplasms"[tiab] OR "pancreas cancer"[tiab] OR "pancreas cancers"[tiab] OR "pancreas neoplasm"[tiab] OR "pancreas neoplasms"[tiab] OR "biliary cancer"[tiab] OR "biliary cancers"[tiab] OR "biliary neoplasm"[tiab] OR "biliary neoplasms"[tiab] OR "biliary carcinoma"[tiab] OR "biliary carcinomas"[tiab] OR "bile duct cancer"[tiab] OR "bile duct cancers"[tiab] OR "bile duct neoplasm"[tiab] OR "bile duct neoplasms"[tiab] OR "biliary tract neoplasm"[tiab] OR "biliary tract neoplasms"[tiab] OR "biliary tract cancer"[tiab] OR "biliary tract cancers"[tiab] OR “bile duct carcinoma”[tiab] OR “bile duct carcinomas”[tiab] OR "colon cancer"[tiab] OR "colon cancers"[tiab] OR "colon neoplasm"[tiab] OR "colon neoplasms"[tiab] OR "colonic neoplasm"[tiab] OR "colonic neoplasms"[tiab] OR "colonic cancer"[tiab] OR "colonic cancers"[tiab] OR "rectal cancer"[tiab] OR "rectal cancers"[tiab] OR "rectal neoplasm"[tiab] OR "rectal

	neoplasms"[tiab] OR "rectum cancers"[tiab] OR "rectum cancer"[tiab] OR "colorectal cancer"[tiab] OR "colorectal cancers"[tiab] OR "colorectal neoplasm"[tiab] OR "colorectal neoplasms"[tiab] OR carcinoid[tiab] OR carcinoids[tiab] OR "duodenal cancer"[tiab] OR "duodenal cancers"[tiab] OR "duodenal neoplasm"[tiab] OR "duodenum cancer"[tiab] OR "duodenum cancers"[tiab] OR "duodenal neoplasms"[tiab] OR "small bowel cancer"[tiab] OR "small bowel cancers"[tiab] OR "small bowel neoplasm"[tiab] OR "small bowel neoplasms"[tiab] OR "gallbladder cancer"[tiab] OR "gallbladder cancers"[tiab] OR "gallbladder neoplasm"[tiab] OR "gallbladder neoplasms"[tiab] OR "gall bladder cancer"[tiab] OR "gall bladder cancers"[tiab] OR "gall bladder neoplasm"[tiab] OR "gall bladder neoplasms"[tiab] OR "anal cancer"[tiab] OR "anal cancers"[tiab] OR "anal neoplasm"[tiab] OR "anal neoplasms"[tiab] OR "anus cancer"[tiab] OR "anus cancers"[tiab] OR "anus neoplasm"[tiab] OR "anus neoplasms"[tiab] OR "Liver Neoplasms"[Mesh] OR "Cholangiocarcinoma"[Mesh] OR "Pancreatic Neoplasms"[Mesh] OR "Biliary Tract Neoplasms"[Mesh] OR "Colonic Neoplasms"[Mesh] OR "Rectal Neoplasms"[Mesh] OR "Colorectal Neoplasms"[Mesh] OR "Duodenal Neoplasms"[Mesh] OR "Gallbladder Neoplasms"[Mesh] OR "Anus Neoplasms"[Mesh] OR "Carcinoid Tumor"[Mesh] OR "Stomach Neoplasms"[Mesh] OR "Esophageal Neoplasms"[Mesh] OR "Adenocarcinoma Of Esophagus" [Supplementary Concept] OR "Esophageal Squamous Cell Carcinoma"[Mesh] OR "Gastrointestinal Neoplasms"[Mesh])
AND	
Particulate Matter / Air Pollution	(“air pollution”[tiab] OR “Particulate Air Pollutants”[tiab] OR “Particulate Air Pollutant”[tiab] OR “particulate matter”[tiab] OR "particulate matters"[tiab] OR "particular matter"[tiab] OR "particular matters"[tiab] OR “air pollutant”[tiab] OR “air pollutants”[tiab] OR "particle pollutant"[tiab] OR "particle pollutants"[tiab] OR “particle pollution”[tiab] OR “fine PM”[tiab] OR "pm2 5"[tiab] OR pm10[tiab] OR "Air Pollution"[Mesh] OR "Particulate Matter"[Mesh] OR "Air Pollutants"[Mesh])
Limits Applied	Language: English
	Publication date: 01/01/1980 – 12/31/2019; 01/01/2019 – 12/31/2020; 01/01/2020 – 12/31/2021

Notes: The limits for language (English) and publication year (1980 – 2021) were applied to the main search using the filters available in PubMed. The keywords were searched in the title and abstract fields in PubMed (i.e., [tiab]) and the controlled vocabulary terms are indicated with [Mesh]. Phrases were enclosed in quotation marks to force the searching of the exact terms in order presented. No other limits were applied to the searches.

Database: Embase
Vendor: Elsevier
Date of search: July 2, 2019, March 15, 2021, September 7, 2021
Limits used: Source: Embase & Embase Classic; Language: English; Publication year: 1980–2019, 2019–2020, 2020–2021

Concept	Search Terms Used
Gastrointestinal Cancer	('liver cancer'/exp OR 'liver tumor'/exp OR 'bile duct carcinoma'/exp OR 'bile duct cancer'/exp OR 'pancreas tumor'/exp OR 'pancreas cancer'/exp OR 'colon cancer'/exp OR 'colon tumor'/exp OR 'colon neoplasms'/exp OR 'rectum cancer'/exp OR 'rectum tumor'/exp OR 'colorectal cancer'/exp OR 'colorectal tumor'/exp OR 'duodenum cancer'/exp OR 'duodenum tumor'/exp OR 'gallbladder cancer'/exp OR 'gallbladder tumor'/exp OR 'anus cancer'/exp OR 'anus tumor'/exp OR 'carcinoid'/exp OR 'stomach cancer'/exp OR 'stomach tumor'/exp OR 'esophagus cancer'/exp OR 'esophagus tumor'/exp OR 'esophageal squamous cell carcinoma'/exp OR 'digestive system cancer'/exp OR 'gastrointestinal tumor'/exp OR "Esophageal cancer":ti,ab OR "Esophageal cancers":ti,ab OR "oesophageal cancer":ti,ab OR "oesophageal cancers":ti,ab OR "Gastric cancer":ti,ab OR "Gastric cancers":ti,ab OR "esophageal adenocarcinoma":ti,ab OR "esophageal adenocarcinomas":ti,ab OR "oesophageal adenocarcinoma":ti,ab OR "oesophageal adenocarcinomas":ti,ab OR "Upper aerodigestive tract cancer":ti,ab OR "Upper aerodigestive tract cancers":ti,ab OR "Stomach cancer":ti,ab OR "Stomach cancers":ti,ab OR "esophageal squamous cell carcinoma":ti,ab OR "esophageal squamous cell carcinomas":ti,ab OR "oesophageal squamous cell carcinoma":ti,ab OR "oesophageal squamous cell carcinomas":ti,ab OR "Upper gastrointestinal cancer":ti,ab OR "Upper gastrointestinal cancers":ti,ab OR "Esophageal neoplasm":ti,ab OR "Esophageal neoplasms":ti,ab OR "oesophageal neoplasm":ti,ab OR "oesophageal neoplasms":ti,ab OR "Gastric neoplasm":ti,ab OR "Gastric neoplasms":ti,ab OR "esophageal adenocarcinoma":ti,ab OR "esophageal adenocarcinomas":ti,ab OR "oesophageal adenocarcinoma":ti,ab OR "oesophageal adenocarcinomas":ti,ab OR "Upper aerodigestive tract neoplasms":ti,ab OR "Stomach neoplasm":ti,ab OR "Stomach neoplasms":ti,ab OR "esophageal squamous cell carcinoma":ti,ab OR "esophageal squamous cell carcinomas":ti,ab OR "oesophageal squamous cell carcinoma":ti,ab OR "oesophageal squamous cell carcinomas":ti,ab OR "Upper gastrointestinal neoplasm":ti,ab OR "Upper gastrointestinal neoplasms":ti,ab OR "alimentary carcinoma":ti,ab OR "gastrointestinal cancer":ti,ab OR "gastrointestinal cancers":ti,ab OR "gastrointestinal neoplasm":ti,ab OR "gastrointestinal neoplasms":ti,ab OR "Gastrointestinal Tract Cancer":ti,ab OR "Gastrointestinal Tract Cancers":ti,ab OR "Gastrointestinal Neoplasms":ti,ab OR "liver cancer":ti,ab OR "liver cancers":ti,ab OR "hepatic neoplasms":ti,ab OR "hepatic neoplasm":ti,ab OR "liver neoplasm":ti,ab OR "liver neoplasms":ti,ab OR "hepatic cancer":ti,ab OR "hepatic cancers":ti,ab OR "hepatic neoplasm":ti,ab OR "hepatic neoplasms":ti,ab OR "hepatocellular cancer":ti,ab OR "hepatocellular cancers":ti,ab OR "hepatocellular neoplasm":ti,ab OR "hepatocellular neoplasms":ti,ab OR cholangiocarcinoma OR cholangiocarcinomas OR "cholangiocellular carcinoma":ti,ab OR "cholangiocellular carcinomas":ti,ab OR "extrahepatic cholangiocarcinoma":ti,ab OR "extrahepatic cholangiocarcinomas":ti,ab OR "intrahepatic cholangiocarcinoma":ti,ab OR "intrahepatic cholangiocarcinomas":ti,ab OR "pancreatic cancer":ti,ab OR "pancreatic cancers":ti,ab OR "pancreatic neoplasm":ti,ab OR "pancreatic neoplasms":ti,ab OR "pancreas cancer":ti,ab OR "pancreas cancers":ti,ab OR "pancreas neoplasm":ti,ab OR "pancreas neoplasms":ti,ab OR "biliary cancer":ti,ab OR "biliary cancers":ti,ab OR "biliary neoplasm":ti,ab OR "biliary neoplasms":ti,ab OR "biliary carcinoma":ti,ab OR "biliary

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AND	
Particulate Matter / Air Pollution	('particulate matter'/exp OR 'air pollutant'/exp OR 'air pollution'/exp OR 'particulate matter 10'/exp OR 'particulate matter 2 5'/exp OR “air pollution”:ti,ab OR “Particulate Air Pollutants”:ti,ab OR “Particulate Air Pollutant”:ti,ab OR “particulate matter”:ti,ab OR "particulate matters":ti,ab OR "particular matter":ti,ab OR "particular matters":ti,ab OR “air pollutant”:ti,ab OR “air pollutants”:ti,ab OR "particle pollutant":ti,ab OR "particle pollutants":ti,ab OR “particle pollution”:ti,ab OR “fine PM”:ti,ab OR "pm2 5":ti,ab OR pm10:ti,ab)
Limits Applied	Language: English
	Publication year: 1980–2019; 2019–2020; 2020–2021
	Source: Embase and Embase Classic

Notes: The limits for language (English) and publication year (1980 – 2021) were applied to the main search using the filters available in Embase. The keywords were searched in the title and abstract fields (i.e., :ti,ab), the controlled vocabulary terms (i.e., EMTREE) are indicated with /exp, and source was selected to limit to the Embase and Embase Classic databases. Phrases were enclosed in quotation marks to force the searching of the exact terms in order presented. No other limits were applied to the searches.

Database: Scopus
Vendor: Elsevier
Date of searches: July 2, 2019, March 15, 2021, September 7, 2021
Limits used: Language: English; Publication year: 1980–2019, 2019–2020, 2020–2021

Concept	Search Terms Used
Gastrointestinal Cancer	TITLE-ABS-KEY(({Esophageal cancer} OR {Esophageal cancers} OR {oesophageal cancer} OR {oesophageal cancers} OR {Gastric cancer} OR {Gastric cancers} OR {esophageal adenocarcinoma} OR {esophageal adenocarcinomas} OR {oesophageal adenocarcinoma} OR {oesophageal adenocarcinomas} OR {Upper aerodigestive tract cancer} OR {Upper aerodigestive tract cancers} OR {Stomach cancer} OR {Stomach cancers} OR {esophageal squamous cell carcinoma} OR {esophageal squamous cell carcinomas} OR {oesophageal squamous cell carcinoma} OR {oesophageal squamous cell carcinomas} OR {Upper gastrointestinal cancer} OR {Upper gastrointestinal cancers} OR {Esophageal neoplasm} OR {Esophageal neoplasms} OR {oesophageal neoplasm} OR {oesophageal neoplasms} OR {Gastric neoplasm} OR {Gastric neoplasms} OR {esophageal adenocarcinoma} OR {esophageal adenocarcinomas} OR {oesophageal adenocarcinoma} OR {oesophageal adenocarcinomas} OR {Upper aerodigestive tract neoplasms} OR {Stomach neoplasm} OR {Stomach neoplasms} OR {esophageal squamous cell carcinoma} OR {esophageal squamous cell carcinomas} OR {oesophageal squamous cell carcinoma} OR {oesophageal squamous cell carcinomas} OR {Upper gastrointestinal neoplasm} OR {Upper gastrointestinal neoplasms} OR {alimentary carcinoma} OR {alimentary carcinomas} OR {gastrointestinal cancer} OR {gastrointestinal cancers} OR {gastrointestinal neoplasm} OR {gastrointestinal neoplasms} OR {Gastrointestinal Tract cancer} OR {Gastrointestinal Tract cancers} OR {Gastrointestinal neoplasms} OR {liver cancer} OR {liver cancers} OR {hepatic neoplasms} OR {hepatic neoplasm} OR {liver neoplasm} OR {liver neoplasms} OR {hepatic cancer} OR {hepatic cancers} OR {hepatic neoplasm} OR {hepatic neoplasms} OR {hepatocellular cancer} OR {hepatocellular cancers} OR {hepatocellular neoplasm} OR {hepatocellular neoplasms} OR cholangiocarcinoma OR cholangiocarcinomas OR {cholangiocellular carcinoma} OR {alimentary carcinoma} OR {cholangiocellular carcinomas} OR {extrahepatic cholangiocarcinoma} OR {extrahepatic cholangiocarcinomas} OR {intrahepatic cholangiocarcinoma} OR {intrahepatic cholangiocarcinomas} OR {pancreatic cancer} OR {pancreatic cancers} OR {pancreatic neoplasm} OR {pancreatic neoplasms} OR {pancreas cancer} OR {pancreas cancers} OR {pancreas neoplasm} OR {pancreas neoplasms} OR {biliary cancer} OR {biliary cancers} OR {biliary neoplasm} OR {biliary neoplasms} OR {biliary carcinoma} OR {biliary carcinomas} OR {bile duct cancer} OR {bile duct cancers} OR {bile duct neoplasm} OR {bile duct neoplasms} OR {biliary tract neoplasm} OR {biliary tract neoplasms} OR {biliary tract cancer} OR {biliary tract cancers} OR {bile duct carcinoma} OR {bile duct carcinomas} OR {colon cancer} OR {colon cancers} OR {colon neoplasm} OR {colon neoplasms} OR {colonic neoplasm} OR {colonic neoplasms} OR {colonic cancer} OR {colonic cancers} OR {rectal cancer} OR {rectal cancers} OR {rectal neoplasm} OR {rectal neoplasms} OR {rectum cancers} OR {rectum cancer} OR {colorectal cancer} OR {colorectal cancers} OR {colorectal neoplasm} OR {colorectal neoplasms} OR carcinoid OR carcinoids OR {duodenal cancer} OR {duodenal cancers} OR {duodenal neoplasm} OR {duodenum cancer} OR {duodenum cancers} OR {duodenal neoplasms} OR {small bowel cancer} OR {small bowel cancers} OR {small bowel neoplasm} OR {small bowel neoplasms}

	OR {gallbladder cancer} OR {gallbladder cancers} OR {gallbladder neoplasm} OR {gallbladder neoplasms} OR {gall bladder cancer} OR {gall bladder cancers} OR {gall bladder neoplasm} OR {gall bladder neoplasms} OR {anal cancer} OR {anal cancers} OR {anal neoplasm} OR {anal neoplasms} OR {anus cancer} OR {anus cancers} OR {anus neoplasm} OR {anus neoplasms} OR {Adenocarcinoma Of Esophagus})
AND	
Particulate Matter / Air Pollution	TITLE-ABS-KEY ({air pollution} OR {Particulate Air Pollutants} OR {Particulate Air Pollutant} OR {particulate matter} OR {particulate matters} OR {particular matter} OR {particular matters} OR {air pollutant} OR {air pollutants} OR {particle pollutant} OR {particle pollutants} OR {particle pollution} OR {fine PM} OR {pm2 5} OR pm10))
Limits Applied	Language: English
	Publication year: 1980–2019; 2019–2020; 2020–2021

Notes: The limits for language (English) and publication year (1980 – 2021) were applied to the main search using the filters available in Scopus. The keywords were searched in the title, abstract, and keyword fields (i.e., TITLE-ABS-KEY). Phrases were enclosed in the curly brackets (i.e., {particulate matter}) to force the searching of the exact terms in order presented. No other limits were applied to the searches.

Databases: Web of Science: Core Collection, MEDLINE, Russian Citation Index, SciELO Citation Index
Vendor: Clarivate Analytics
Date of searches: July 2, 2019, March 15, 2021, September 7, 2021
Limits used: Language: English; Publication year: 1980–2019, 2019–2020, 2020–2021

Concept	Search Terms Used
Gastrointestinal Cancer	TS=(("Esophageal cancer" OR "Esophageal cancers" OR "oesophageal cancer" OR "oesophageal cancers" OR "Gastric cancer" OR "Gastric cancers" OR "esophageal adenocarcinoma" OR "esophageal adenocarcinomas" OR "oesophageal adenocarcinoma" OR "oesophageal adenocarcinomas" OR "Upper aerodigestive tract cancer" OR "Upper aerodigestive tract cancers" OR "Stomach cancer" OR "Stomach cancers" OR "esophageal squamous cell carcinoma" OR "esophageal squamous cell carcinomas" OR "oesophageal squamous cell carcinoma" OR "oesophageal squamous cell carcinomas" OR "Upper gastrointestinal cancer" OR "Upper gastrointestinal cancers" OR "Esophageal neoplasm" OR "Esophageal neoplasms" OR "oesophageal neoplasm" OR "oesophageal neoplasms" OR "Gastric neoplasm" OR "Gastric neoplasms" OR "esophageal adenocarcinoma" OR "esophageal adenocarcinomas" OR "oesophageal adenocarcinoma" OR "oesophageal adenocarcinomas" OR "Upper aerodigestive tract neoplasms" OR "Stomach neoplasm" OR "Stomach neoplasms" OR "esophageal squamous cell carcinoma" OR "esophageal squamous cell carcinomas" OR "oesophageal squamous cell carcinoma" OR "oesophageal squamous cell carcinomas" OR "Upper gastrointestinal neoplasm" OR "Upper gastrointestinal neoplasms" OR "alimentary carcinoma" OR "gastrointestinal cancer" OR "gastrointestinal cancers" OR "gastrointestinal neoplasm" OR "gastrointestinal neoplasms" OR "Gastrointestinal Tract Cancer" OR "Gastrointestinal Tract Cancers" OR "Gastrointestinal Neoplasms" OR "liver cancer" OR "liver cancers" OR "hepatic neoplasms" OR "hepatic neoplasm" OR "liver neoplasm" OR "liver neoplasms" OR "hepatic cancer" OR "hepatic cancers" OR "hepatic neoplasm" OR "hepatic neoplasms" OR "hepatocellular cancer" OR "hepatocellular cancers" OR "hepatocellular neoplasm" OR "hepatocellular neoplasms" OR cholangiocarcinoma OR cholangiocarcinomas OR "cholangiocellular carcinoma" OR "cholangiocellular carcinomas" OR "extrahepatic cholangiocarcinoma" OR "extrahepatic cholangiocarcinomas" OR "intrahepatic cholangiocarcinoma" OR "intrahepatic cholangiocarcinomas" OR "pancreatic cancer" OR "pancreatic cancers" OR "pancreatic neoplasm" OR "pancreatic neoplasms" OR "pancreas cancer" OR "pancreas cancers" OR "pancreas neoplasm" OR "pancreas neoplasms" OR "biliary cancer" OR "biliary cancers" OR "biliary neoplasm" OR "biliary neoplasms" OR "biliary carcinoma" OR "biliary carcinomas" OR "bile duct cancer" OR "bile duct cancers" OR "bile duct neoplasm" OR "bile duct neoplasms" OR "biliary tract neoplasm" OR "biliary tract neoplasms" OR "biliary tract cancer" OR "biliary tract cancers" OR “bile duct carcinoma” OR “bile duct carcinomas” OR "colon cancer" OR "colon cancers" OR "colon neoplasm" OR "colon neoplasms" OR "colonic neoplasm" OR "colonic neoplasms" OR "colonic cancer" OR "colonic cancers" OR "rectal cancer" OR "rectal cancers" OR "rectal neoplasm" OR "rectal neoplasms" OR "rectum cancers" OR "rectum cancer" OR "colorectal cancer" OR "colorectal cancers" OR "colorectal neoplasm" OR "colorectal neoplasms" OR carcinoid OR carcinoids OR "duodenal cancer" OR "duodenal cancers" OR "duodenal neoplasm" OR "duodenum cancer" OR "duodenum cancers" OR "duodenal neoplasms" OR "small bowel cancer" OR "small bowel cancers" OR "small bowel neoplasm" OR "small bowel neoplasms" OR "gallbladder cancer" OR "gallbladder cancers" OR "gallbladder neoplasm" OR

	"gallbladder neoplasms" OR "gall bladder cancer" OR "gall bladder cancers" OR "gall bladder neoplasm" OR "gall bladder neoplasms" OR "anal cancer" OR "anal cancers" OR "anal neoplasm" OR "anal neoplasms" OR "anus cancer" OR "anus cancers" OR "anus neoplasm" OR "anus neoplasms" OR "Adenocarcinoma Of Esophagus"))
AND	
Particulate Matter / Air Pollution	TS=((“air pollution” OR “Particulate Air Pollutants” OR “Particulate Air Pollutant” OR “particulate matter” OR "particulate matters" OR "particular matter" OR "particular matters" OR “air pollutant” OR “air pollutants” OR "particle pollutant" OR "particle pollutants" OR “particle pollution” OR “fine PM” OR "pm2 5" OR pm10))
Limits Applied	Language: English
	Publication year: 1980–2019, 2019–2020, 2020–2021

Notes: The limits for language (English) and publication year (1980 – 2021) were applied to the main search using the filters available in Web of Science. Search “All Databases” then select “advanced search” and uncheck the databases to not search. Then search using the Topic field (i.e., TS) to search the title, abstract, keywords, Keywords Plus fields. Phrases were enclosed in quotation marks to force the searching of the exact terms in order presented. No other limits were applied to the searches.
Web of Science: Core Collection includes the following databases available through our library subscription: Science Citation Index Expanded (SCI-EXPANDED)--1900-present; Social Sciences Citation Index (SSCI)--1900-present; Conference Proceedings Citation Index – Science (CPCI-S)--1990-present; Conference Proceedings Citation Index – Social Science & Humanities (CPCI-SSH)--1990-present; Book Citation Index – Science (BKCI-S)--2005-present; Book Citation Index – Social Sciences & Humanities (BKCI-SSH)--2005-present; Emerging Sources Citation Index (ESCI)--2005-present; Current Chemical Reactions (CCR-EXPANDED)--1985-present.

Database: Cochrane Library
Vendor: Wiley & Sons
Date of searches: July 2, 2019, March 15, 2021, September 7, 2021
Limits used: Language: English; Publication year: 1980–2019, 2019–2020, 2020–2021

Concept	Set	Search Terms Used
Gastrointestinal Cancer	#1	("Esophageal cancer" OR "Esophageal cancers" OR "oesophageal cancer" OR "oesophageal cancers" OR "Gastric cancer" OR "Gastric cancers" OR "esophageal adenocarcinoma" OR "esophageal adenocarcinomas" OR "oesophageal adenocarcinoma" OR "oesophageal adenocarcinomas" OR "Upper aerodigestive tract cancer" OR "Upper aerodigestive tract cancers" OR "Stomach cancer" OR "Stomach cancers" OR "esophageal squamous cell carcinoma" OR "esophageal squamous cell carcinomas" OR "oesophageal squamous cell carcinoma" OR "oesophageal squamous cell carcinomas" OR "Upper gastrointestinal cancer" OR "Upper gastrointestinal cancers" OR "Esophageal neoplasm" OR "Esophageal neoplasms" OR "oesophageal neoplasm" OR "oesophageal neoplasms" OR "Gastric neoplasm" OR "Gastric neoplasms" OR "esophageal adenocarcinoma" OR "esophageal adenocarcinomas" OR "oesophageal adenocarcinoma" OR "oesophageal adenocarcinomas" OR "Upper aerodigestive tract neoplasms" OR "Stomach neoplasm" OR "Stomach neoplasms" OR "esophageal squamous cell carcinoma" OR "esophageal squamous cell carcinomas" OR "oesophageal squamous cell carcinoma" OR "oesophageal squamous cell carcinomas" OR "Upper gastrointestinal neoplasm" OR "Upper gastrointestinal neoplasms" OR "alimentary carcinoma" OR "gastrointestinal cancer" OR "gastrointestinal cancers" OR "gastrointestinal neoplasm" OR "gastrointestinal neoplasms" OR "Gastrointestinal Tract Cancer" OR "Gastrointestinal Tract Cancers" OR "Gastrointestinal Neoplasms" OR "liver cancer" OR "liver cancers" OR "hepatic neoplasms" OR "hepatic neoplasm" OR "liver neoplasm" OR "liver neoplasms" OR "hepatic cancer" OR "hepatic cancers" OR "hepatic neoplasm" OR "hepatic neoplasms" OR "hepatocellular cancer" OR "hepatocellular cancers" OR "hepatocellular neoplasm" OR "hepatocellular neoplasms" OR cholangiocarcinoma OR cholangiocarcinomas OR "cholangiocellular carcinoma" OR "cholangiocellular carcinomas" OR "extrahepatic cholangiocarcinoma" OR "extrahepatic cholangiocarcinomas" OR "intrahepatic cholangiocarcinoma" OR "intrahepatic cholangiocarcinomas" OR "pancreatic cancer" OR "pancreatic cancers" OR "pancreatic neoplasm" OR "pancreatic neoplasms" OR "pancreas cancer" OR "pancreas cancers" OR "pancreas neoplasm" OR "pancreas neoplasms" OR "biliary cancer" OR "biliary cancers" OR "biliary neoplasm" OR "biliary neoplasms" OR "biliary carcinoma" OR "biliary carcinomas" OR "bile duct cancer" OR "bile duct cancers" OR "bile duct neoplasm" OR "bile duct neoplasms" OR "biliary tract neoplasm" OR "biliary tract neoplasms" OR "biliary tract cancer" OR "biliary tract cancers" OR “bile duct carcinoma” OR “bile duct carcinomas” OR "colon cancer" OR "colon cancers" OR "colon neoplasm" OR "colon neoplasms" OR "colonic neoplasm" OR "colonic neoplasms" OR "colonic cancer" OR "colonic cancers" OR "rectal cancer" OR "rectal cancers" OR "rectal neoplasm" OR "rectal neoplasms" OR "rectum cancers" OR "rectum cancer" OR "colorectal cancer" OR "colorectal cancers" OR "colorectal neoplasm" OR "colorectal neoplasms" OR carcinoid OR carcinoids OR "duodenal cancer"

		OR "duodenal cancers" OR "duodenal neoplasm" OR "duodenum cancer" OR "duodenum cancers" OR "duodenal neoplasms" OR "small bowel cancer" OR "small bowel cancers" OR "small bowel neoplasm" OR "small bowel neoplasms" OR "gallbladder cancer" OR "gallbladder cancers" OR "gallbladder neoplasm" OR "gallbladder neoplasms" OR "gall bladder cancer" OR "gall bladder cancers" OR "gall bladder neoplasm" OR "gall bladder neoplasms" OR "anal cancer" OR "anal cancers" OR "anal neoplasm" OR "anal neoplasms" OR "anus cancer" OR "anus cancers" OR "anus neoplasm" OR "anus neoplasms"):ti,ab,kw
Gastrointestinal Cancer	#2	([mh "Liver Neoplasms"] OR [mh Cholangiocarcinoma] OR [mh "Pancreatic Neoplasms"] OR [mh "Biliary Tract Neoplasms"] OR [mh "Colonic Neoplasms"] OR [mh "Rectal Neoplasms"] OR [mh "Colorectal Neoplasms"] OR [mh "Duodenal Neoplasms"] OR [mh "Gallbladder Neoplasms"] OR [mh "Anus Neoplasms"] OR [mh "Carcinoid Tumor"] OR [mh "Stomach Neoplasms"] OR [mh "Esophageal Neoplasms"] OR [mh "Adenocarcinoma Of Esophagus"] OR [mh "Esophageal Squamous Cell Carcinoma"] OR [mh "Gastrointestinal Neoplasms"])
Particulate Matter /Air Pollution	#3	(“air pollution” OR “Particulate Air Pollutants” OR “Particulate Air Pollutant” OR “particulate matter” OR "particulate matters" OR "particular matter" OR "particular matters" OR “air pollutant” OR “air pollutants” OR "particle pollutant" OR "particle pollutants" OR “particle pollution” OR “fine PM” OR "pm2 5" OR pm10):ti,ab,kw
Particulate Matter /Air Pollution	#4	([mh "Air Pollution"] OR [mh "Particulate Matter"] OR [mh "Air Pollutants"])
	#5	#1 OR #2
	#6	#3 OR #4
	#7	#5 AND #6
Limits Applied		Language: English
		Publication year: 1980–2019, 2019–2020, 2020–2021

Notes: The limits for language (English) and publication year (1980 – 2021) were applied to the main search using the limits available in the Cochrane Library database. The keywords were searched in the title, abstract, and keyword fields (i.e., :ti,ab,kw) and the controlled vocabulary terms (i.e., MeSH) are indicated with [mh]. Phrases were enclosed in quotation marks to force the searching of the exact terms in order presented. No other limits were applied to the searches.

Table S3. Detail and rationale for risk of bias ratings for included human studies relevant to PM exposure and GI cancer outcomes

Article No.	Study Author	Study group representation	Outcome assessment methods	Subtypes?	Exposure assessment methods	Confounding	Statistical analysis	Conflict of interest	Overall rating
1	Ancona et al.	Large study population but only in Rome	Individual level incidence	No	Point source data collected mid-study with nonstandard units; no characteristics of $PM_{2.5}$	Age, gender, region of birth, educational, occupation, marital status, and an area-based socio-economic position index; stratified analyses conducted by gender	Multiple comparisons with no adjustments; limited sensitivity analyses	Included statement of funding or COI	Probably high risk
2	Bogumil et al.	Large, diverse study population in California	Individual level incidence from SEER linkage	Yes	U.S. EPA routine air monitoring stations and location of residence using Kriging spatial interpolation and standard units	Race/ethnicity, BMI, neighborhood SES, age at cohort entry, sex, diabetes status, BMI, smoking status, work history, Alternative Healthy Eating Index 2010, alcohol consumption	Models with the first 5 years of follow-up time removed from analysis; stratified by sex, race/ethnicity, smoking status, moving status, BMI, and neighborhood SES; inclusion of quartile categories and quadratic terms of the air pollutants	Included statement of funding or COI	Low
3	Chu et al.	Large study population from the PLCO Cancer Screening Trial, a population-based cohort in the US	Diagnosis of colorectal cancer was histologically confirmed via medical record reviews, the National	No	Individual mean $PM_{2.5}$ estimates from monitoring data collected from the date of trial entry to the date of CRC diagnosis or trial exit; no co-	Sex, age, study arm, smoking status, occupation status, educational level, and BMI	Both linear and non-linear relationship between $PM_{2.5}$ and CRC was examined using natural splines with two degrees of freedom;	Included statement of funding or COI	Low

			Death Index, and self-reported annual questionnaires during follow-up		pollutants; non-standard units		stratified analyses were conducted by median $PM_{2.5}$ value		
4	Coleman et al.	Large NHIS study	County level mortality data	No	Census tract level data; back casted estimates for first 12 years of study; no co-pollutants or characteristics of $PM_{2.5}$	Age-sex-race interactions (using indicators for 5-year age buckets) and categorical variables for BMI, income, education, marital status, smoking status, rural versus urban, region, and survey year stratified analyses by never and ever smokers	Stratification by age, sex, and race/ethnicity; Holms adjusted p values to account for multiple testing	Included statement of funding or COI	Probably low risk
5	Coleman et al.	Large SEER study	County level incidence	No	Only county level, not available for first 5 years; back casted estimates; no co-pollutants or characteristics of $PM_{2.5}$	County level proportions for sex, race/ethnicity, 5 year age categories, education, median income, median home value or rent, percent below poverty, percent unemployed, working class, urban or rural, smoking, alcohol consumption, physical activity, obesity, state and region	Lag and spline analyses, but no stratification; Holm's method and Bonferroni correction to adjust for multiple comparisons	Included statement of funding or COI	Probably low risk
6	Datzman n et al.	Large study population but only Saxony and no farmers	Statutory health insurance linked data	No	Only available for one year and linked to postal code; no characteristics of $PM_{2.5}$	Individual level age, sex, alcohol-related disorder, number of physician contacts, and proportion of short and long term unemployment; postal code level information adjustments made for population density, all-cause mortality, and proportion of individuals with alcohol-related disorders	Sensitivity analysis for change of residence	Included statement of funding or COI	Probably high risk
7	Deng et al.	Large study population but only California	California cancer mortality registry	Yes	Residence linked monitor data for duration of study; no co-pollutants or characteristics of $PM_{2.5}$; nonstandard units	Adjusted for age, sex, race/ethnicity, marital status, year of diagnosis, month of diagnosis, initial treatment, SES, metropolitan versus not, and distance to primary highways	Model check for non-linear exposure; stratified by tumor stage at diagnosis, distance to monitor, geocoding	Included statement of funding or COI	Probably low risk

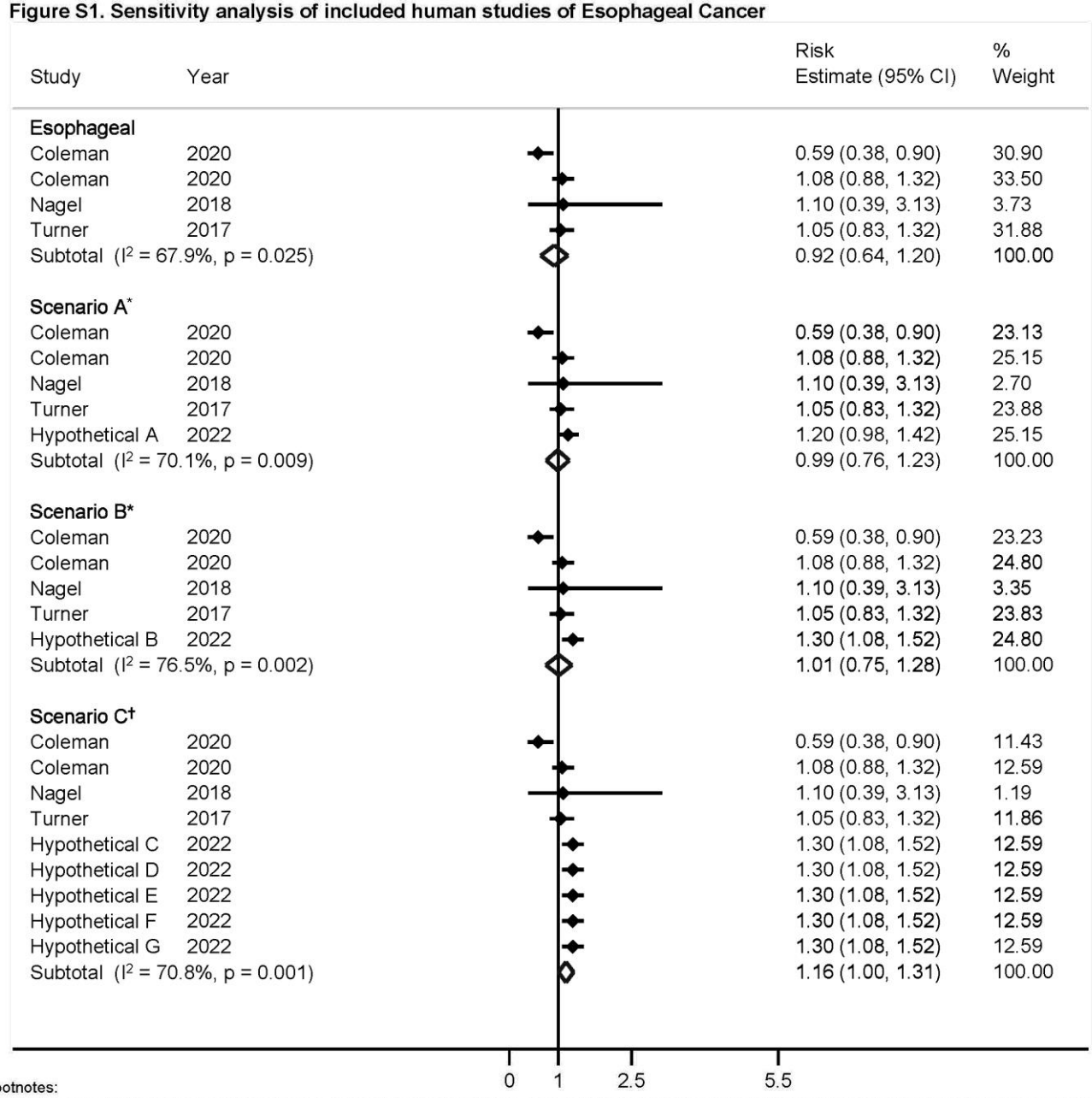
							accuracy, regions, and SES		
8	Ethan et al.	Large population, but only six districts (Beilin, Yanta, Weiyang, Lianhu, Gaoling, and Huyi) within Xi'an	City level mortality	No	City-wide average data from monitoring stations collected throughout the duration of the study; no characteristics of $PM_{2.5}$	Sex, age, date of birth, tribe, marital status, home address, incidence date, death date, morphology (based on ICD-03), behavior, grades, diagnostic basis, ICD-10 codes, last contact time/follow-up deadline, status, and cause of death	Lag and spline analyses; stratification by sex and age (over or under 65)	Included statement of funding or COI	Probably high risk
9	Guo et al.	Large study population but only high SES Taiwan	Individual level mortality	No	Residence linked satellite data for duration of study; no co-pollutants or characteristics of $PM_{2.5}$	Age, sex, education level, body mass index, cigarette smoking, alcohol drinking, physical activity, vegetable or fruit intake, occupational exposure to dust or solvent, season, and year of enrollment	Modifying effects assessed by stratified analyses by sex, age, and smoking, but no adjustment for multiple comparisons	Included statement of funding or COI	Probably low risk
10	Jerrett et al.	Large study population but only LA	County level mortality	No	Zip code linked stationary monitor data for last year of study only; no characteristics of $PM_{2.5}$	Adjusted for 44 confounders related to lifestyle, dietary, demographic, occupational, educational factors, O_3 co-pollutant	Limited sensitivity analyses and no adjustment for multiple comparisons	Included statement of funding or COI	Probably high risk
11	Ma, et al	Large study population, but only subset of diabetic patients	Incidence from National Health Insurance Research Database with ICD-9	Yes	Monitoring data for duration of study was linked to residential address through kriging; co-pollutants were included;	Age, gender, income and level of urbanization; associated comorbidities, with a diagnosis validated by at least one hospital admission or more than three outpatient visits within 1 year; Prescription medications that could potentially confound the association between diabetes and cancer risks	Modifying effects assessed by stratified analyses; model check for non-linear exposure	Included statement of funding or COI	Low Risk

			codes		standard units used				
12	Nagel et al.	Large cohort from 11 countries	Registry data linkage of individual incidence	No	Long-term exposure modeled from monitoring data collected several years after study period; no characteristics of $PM_{2.5}$; non-standard units	Age, calendar year, sex, smoking status, smoking intensity, smoking duration, occupational exposure, employment status, educational level. sensitivity analyses for stable residence, urban versus rural, alcohol consumption, environmental tobacco smoke, intake of fruit, intake of meat, marital status, nitrogen dioxide, nitrogen oxides, and traffic intensity	Sensitivity analysis for change of residence; model check for non-linear exposure	Included statement of funding or COI	Low Risk
13	Pan et al.	Large study population but only seven townships within Taiwan	Registry data linkage of individual incidence	Yes	$PM_{2.5}$ was unavailable for first 15 years of study; no co-pollutants or characteristics of $PM_{2.5}$; non-standard units	Age, sex, alcohol consumption, smoking status, ALT levels, serostatus of HBsAg and anti-HCV	Modifying effects assessed by stratified analyses; model check for non-linear exposure	Included statement of funding or COI	Probably low risk
14	Pedersen et al.	Multi-site study	Linkage to population-based cancer registries with individual incidence data	Yes	Long-term exposure modeled from monitoring data collected several years after study period; no characteristics of $PM_{2.5}$; non-standard units	Age, calendar year, sex, smoking status (never/former/current), alcohol consumption, ever worked as a waiter or cook (high risk occupations for liver cancer), employment status, educational level, area-level SES, stable residential address, nitrogen dioxide, nitrogen oxides, traffic density (vehicles per day), and traffic intensity (major roads)	Sensitivity analysis for change of residence; modifying effects assessed by stratified analyses; model check for non-linear exposure	Included statement of funding or COI	Low Risk
15	So et al.	Multicenter study	Individual cancer diagnosis data from national and state cancer registries; one cohort	Yes	Europe-wide hybrid LUR models at a fine spatial scale (100m x 100m grids) to estimate annual mean exposure for the	Age, sex (strata), subcohort (strata), cohort baseline year, smoking (never, former, current), employment status (employed, other), and neighborhood SES (mean income in 2001)	Residential history was incorporated, stratified analyses for sex and cohort, applied natural cubic splines with	Included statement of funding or COI	Low Risk

			used biannual questionnaire self-reports confirmed by oncologist review of pathological reports or death certificates		year 2010 at the participants' residential address at baseline; non-standard units		two degrees of freedom, lag analysis using strata per year or 5-year of follow-up		
16	Turner et al.	Large study, but mostly friends and family of study volunteers	Individual level mortality	No	Only last 5 years of study; no characteristics of $PM_{2.5}$; non-standard units	Adjusted for education, marital status, BMI, BMI^2, smoking, usual dietary intake of vegetables, fruit, fiber, fat, alcohol, occupational exposures	Modifying effects assessed by stratified analyses by age categories, gender, race/ethnicity, but no adjustment for multiple comparisons	Included statement of funding or COI	Probably low risk
17	VoPham et al.	Large sample size	County level incidence	Yes	County level exposure for year 2000 only	County level risk factors, so no control for individual level smoking; but included county level data on age, sex, race, year, SEER registry, urbanicity, heavy alcohol use, smoking, obesity, diabetes, SES, foreign born, ambient UV exposure, population density	Lag and spline analyses; sensitivity analysis for change of residence; stratified analyses by county, age, sex, race, study year	Included statement of funding or COI	Low Risk
18	Wang et al.	Large sample size	Area level mortality	Yes	Area level estimates for 5 years during study period; no co-pollutants or characteristics of	Few area level risk factors, no SES or smoking, but included adjustment for spatial variation in gender, urban/rural status, age, period, and cohort effect	Model check for non-linear exposure	Included statement of funding or COI	Probably high risk

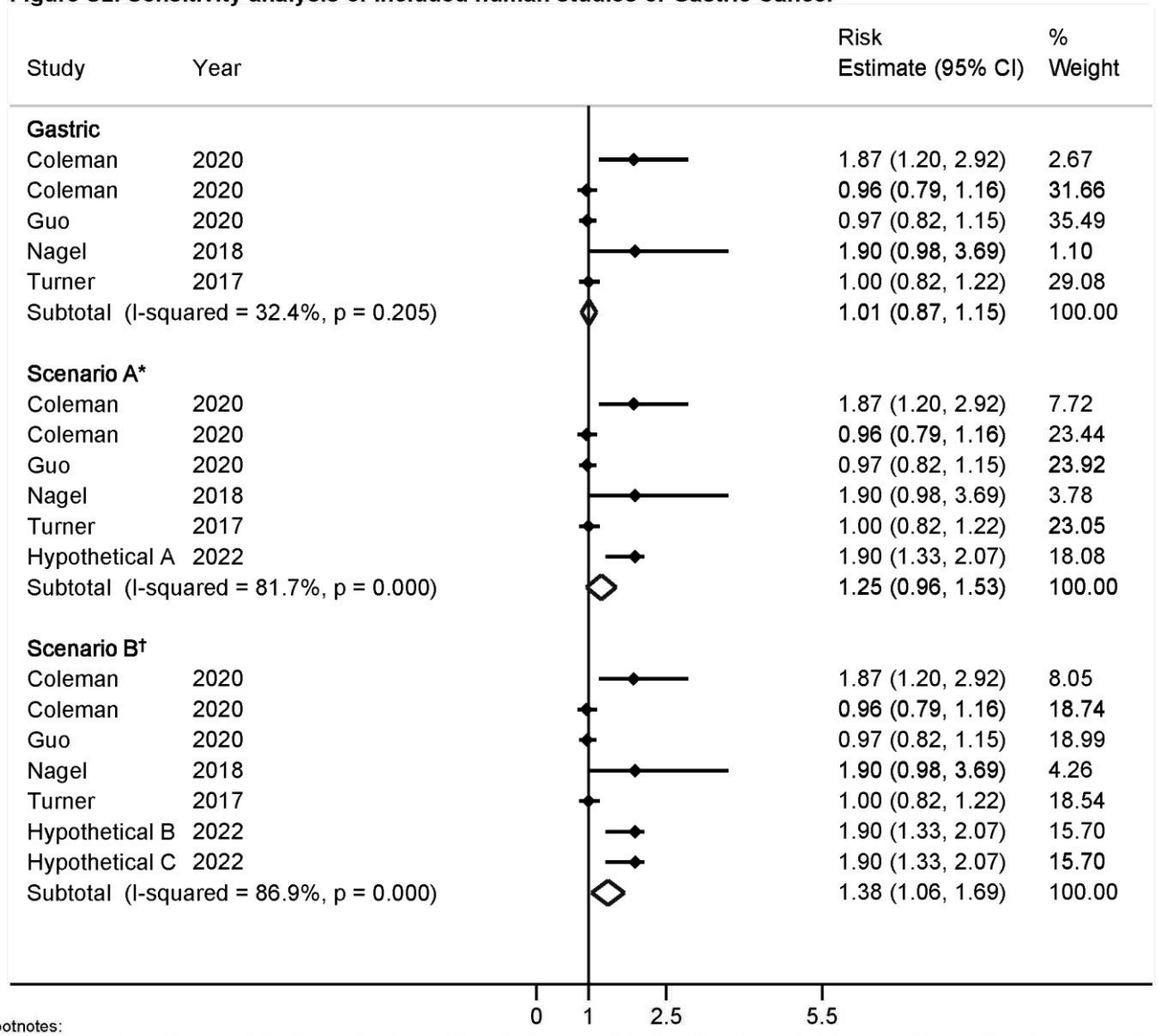
					$PM_{2.5}$				
19	Weinmayr	Large cohort from 7 sites across 6 countries	Linkage to population-based cancer registries with individual incidence data	No	$PM_{2.5}$ components measured after recruitment of cohort participants; non-standard units	Age, calendar year, sex, smoking, occupational exposure, employment status, education, area-level SES; alcohol, environmental tobacco smoke, fruit and meat intake, marital status, urban/rural indicator	Model check for non-linear exposure; restricted analyses to participants with stable residence during follow-up	Included statement of funding or COI	Low Risk
20	Wong et al.	Large sample size, but only over 65 years old within Hong Kong	Hong Kong death registry	No	Monitor and satellite-based data for recruitment year only; no co-pollutants or characteristics of $PM_{2.5}$	Age, gender, BMI, smoking status, exercise frequency, education, and personal monthly expenditure; census tract level data on % older subjects, % tertiary education, and monthly domestic household income; district level data on % smokers and ad hoc survey of ground radon	Spline, change of residence, and stratified analyses, but no adjustment for multiple comparisons	Included statement of funding or COI	Probably low risk

Figure S1. Sensitivity analysis of included human studies of Esophageal Cancer



Footnotes:
*Scenarios demonstrating the magnitude of association that would need to be reported by a hypothetical new study to reverse the overall direction of association.
† Scenario demonstrating the number of new studies of the same magnitude that would be needed for the overall findings to become statistically significant.

Figure S2. Sensitivity analysis of included human studies of Gastric Cancer

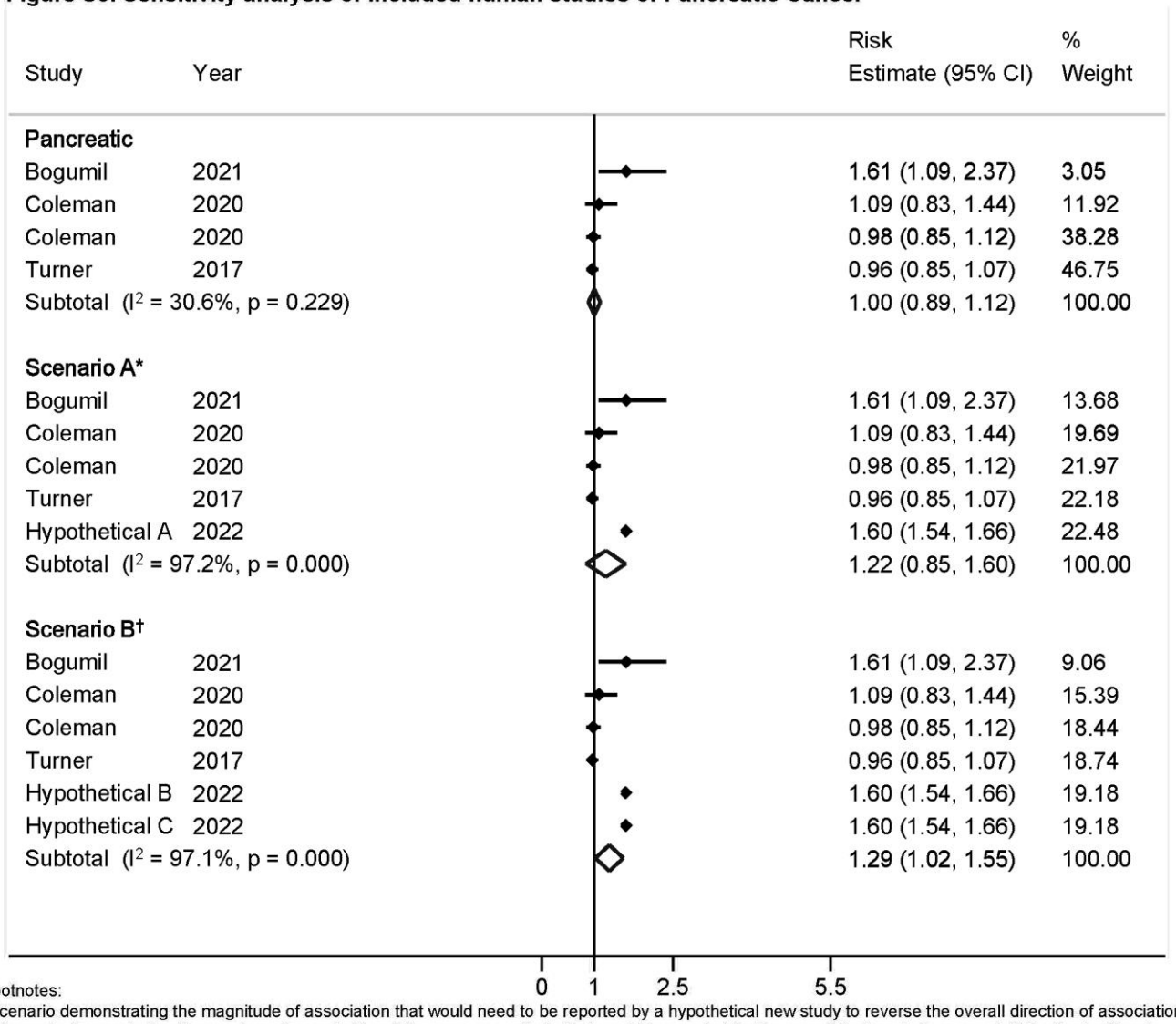


Footnotes:

*Scenario demonstrating the magnitude of association that would need to be reported by a hypothetical new study to reverse the overall direction of association.

† Scenario demonstrating the number of new studies of the same magnitude that would be needed for the overall findings to become statistically significant.

Figure S3. Sensitivity analysis of included human studies of Pancreatic Cancer



Footnotes:

*Scenario demonstrating the magnitude of association that would need to be reported by a hypothetical new study to reverse the overall direction of association.

† Scenario demonstrating the number of new studies of the same magnitude that would be needed for the overall findings to become statistically significant.

Figure S4. Sensitivity analysis of included human studies of Colorectal cancer

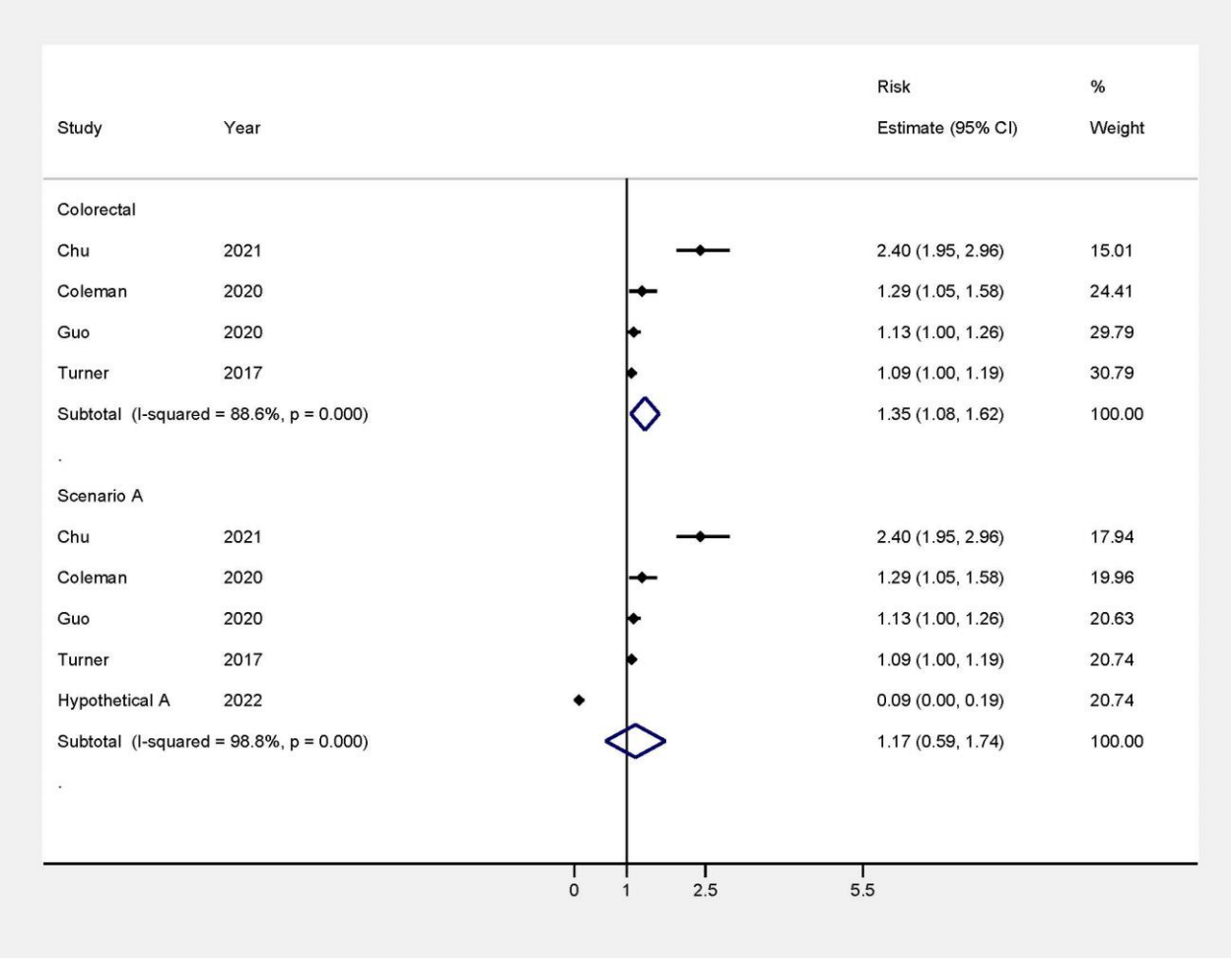


Figure S5. Sensitivity analysis of included human studies of liver cancer

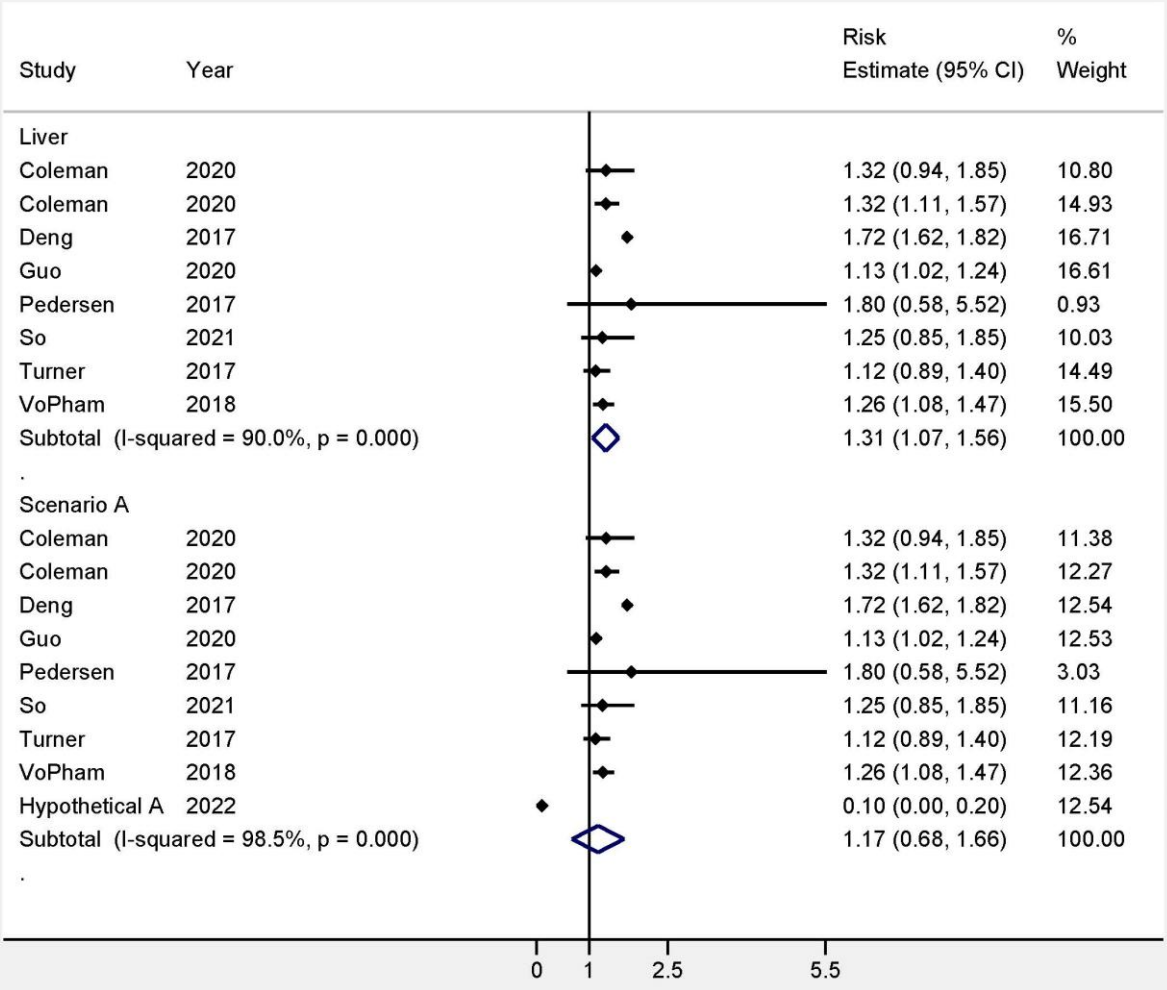


Figure S6. Funnel plot of studies of the association between $PM_{2.5}$ and gastric cancer

